

PROBLEM 3 (1 hour)

A perfect liquid of constant density ρ and constant viscosity μ fills the space between two infinite parallel walls separated by a distance $2h$, that move parallel to themselves in the same direction with a constant velocity U . As shown in the figure, a very thin solid plate of length $L \gg h$ is placed at the center of the channel. In the regions far upstream and far downstream from the plate the liquid moves with uniform velocity U due to the motion of the confining walls, and the modified pressure takes uniform values $P(x \rightarrow -\infty) \rightarrow P_{-\infty}$ and $P(x \rightarrow \infty) \rightarrow P_{\infty}$, respectively.

1. Provide the criterion that must be accomplished for the flow to be dominated by viscous forces. **(1 point)**
2. Obtain the velocity field in the region occupied by the plate, in terms of the unknown pressure gradient $P_l = -\partial P/\partial x$. **(3 points)**
3. Obtain the pressure drop $P_{-\infty} - P_{\infty}$, as well as the pressure gradient P_l . **(3 points)**
4. Determine the forces (per unit spanwise length) that the liquid exerts on the confining walls, F_w , and on the inner plate, F_p , as well as the power that must be applied to drive the walls. **(3 points)**
5. If all the inner walls are thermally isolated, determine the temperature increase in the liquid between two points situated far downstream and far upstream, $T_{\infty} - T_{-\infty}$. To that end, apply the energy equation in integral form to an appropriately defined control volume **(1 extra point)**.

